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Gautam Buddha University
School of Management
Greater Noida
MBA (Business Analytics and Data Science) Programme

Course: Business Intelligence and Analytics Lab

Code:	MBA-472	**Year/Sem: 2024-**2025/II
Department:	School of Management	**Credits:** 02
Sessions:	30	**Each Session: 1 2 0** Minutes

Course Description:

The Descriptive Analytics Lab is a hands-on course designed to provide students with practical experience in applying descriptive analytics techniques to real-world datasets. This lab course emphasizes the use of statistical tools, data visualization techniques, and reporting methods to explore and summarize historical data. Students will work with various datasets to uncover patterns, trends, and insights that inform business decision-making.

In this lab, students will engage in activities such as data cleaning, preparation, and transformation, and apply statistical and visualization methods to analyze past performance. They will learn how to construct meaningful reports and dashboards, summarize data, and create visualizations such as bar charts, pie charts, histograms, line graphs, and dashboards. By using popular analytics tools like IBM Cognos Analytics, students will develop a strong understanding of the practical application of descriptive analytics.

Course Outcome:

- Ability to navigate and perform basic operations within IBM Cognos Analytics, including account creation and navigation.
- Competence in creating various reports, such as list-type reports from external datasets, with emphasis on formatting to enhance readability and visual appeal.
- Application of advanced features like crosstab reports, dynamic/static headers and footers, custom column addition, and drill-through reports to create comprehensive and insightful reports.
- Proficient generation of visualizations and reports using a variety of chart types to analyze data effectively and derive meaningful insights.
- Proficiency in developing active reports with controls, creating stories and dashboards, and modifying report behavior to enhance interactivity and user experience.

Sanjay

Pedagogy:

This pedagogy will make use of analytical and computer programs using Python and other tools having applications mostly in business areas. The sessions will be conducted in Computer lab where students are expected to implement the programs in order to develop programming and problem-solving skills.

Evaluation Scheme: (Practical Paper)

Assignments / Class participation, Machine Test and Presentations	: 50%
End-Sem Exam (Lab Exam)	: 50%

List of Experiments:

1. Account creation
2. Basic operation of IBM Cognos
3. Create list type report from external data sets
4. Create list for showing data, create list type report from external data sets
5. Apply formatting in list and other reports.
6. Create crosstab report by using internal datasets and all types of filters
7. Apply filters on various types of reports.
8. Create visualization, reports using various types of charts
9. Apply various types of prompts on report, create dynamic and static header, footer in report
10. Create page header and footer add custom column in report.
11. Create drill through report
12. Build query models and then connect them to the report layout
13. Filter data using the query model
14. Create a report comparing quantity sold in different order years
15. Create a story from basic to advance
16. Create a dashboard from basic to advance
17. Create a simple active report using static and data-driven controls and change filtering and selection behavior in a report.

Sanjay

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Gautam Buddha University
School of Management
Greater Noida
MBA (Business Analytics and Data Science) Programme
Course: Business Intelligence and Analytics (MBA-462)

Instructor:	Mr. Akshay Taneja	**Session/Sem:**	2024-2025/II
Department:	School of Management	**Credits:**	02
Sessions:	30	**Each Session:**	60 Minutes

Course Description:

Descriptive Analytics is a foundational course designed to provide students with a deep understanding of how to summarize, visualize, and interpret historical data to inform business decision-making. The course covers techniques to analyze past performance and trends through various data visualization tools, statistical measures, and data aggregation methods. Students will explore key concepts such as data cleaning, trend analysis, and basic statistical techniques, helping them build a clear picture of what has happened within a dataset.

The course focuses on practical applications of descriptive analytics in real-world scenarios, teaching students how to generate actionable insights from data. Topics will include the use of charts, graphs, dashboards, and reports, as well as how to identify patterns and anomalies in data. Participants will also gain hands-on experience with tools that are essential for analyzing historical data, such as IBM Cognos Analytics, a powerful platform for visualizing, reporting, and analyzing business data.

IBM Cognos Analytics is an integral part of this course, as it provides robust capabilities for descriptive analytics. It is a comprehensive business intelligence (BI) tool designed to help users create, manage, and share reports and dashboards. IBM Cognos allows for seamless data visualization, which enables users to transform raw data into insightful charts, graphs, and interactive dashboards. The platform's intuitive interface makes it easy to explore historical trends, summarize key metrics, and identify opportunities for optimization.

Course Objectives:

- Acquire a deep understanding of analytics and business intelligence principles.
- Develop proficiency in using IBM Cognos Analytics for data analysis and reporting tasks.
- Master SQL query writing and summarization techniques for efficient data retrieval and analysis.
- Gain insights into the significance of data-driven decision-making across diverse industries.
- Learn to create advanced reports, dashboards, and visualizations to convey insights effectively.
- Explore the application of predictive analytics in transforming organizational strategies and processes.
- Understand the role of analytics in enhancing business performance, managing assets, and combating fraud.

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Learning Outcomes:

After completing the course, the students should be able to:

1. Understand fundamental analytics principles.

2. Utilize IBM Cognos Analytics proficiently for data analysis and reporting.

3. Master SQL query writing for effective data manipulation.

4. Apply data-driven decision-making across industries.

5. Create advanced reports, dashboards, and visualizations.

6. Implement predictive analytics techniques for forecasting and optimization.

7. Apply business intelligence strategies to enhance organizational performance.

Pedagogy: The pedagogy will be a mix of lectures, hands on learning, real life case discussions, assignments and industry/research-based projects. In addition to the reading materials, additional readings and cases will be distributed in the class from time to time. Students are also expected to prepare and analyze all the cases as class participation is very important.

Evaluation Scheme:

➢ Quizzes/ Class Tests/Presentations	15%
➢ Mid-Semester Exam	25%
➢ Term- Semester Exam	60%

Suggested Readings:

Text Book:

1. Liautaud, Bernard, and Mark Hammond. e-Business intelligence: turning information into knowledge into profit. McGraw-Hill, Inc., 2000.

Reference Books:

1. P Anandarajan, Murugan, Asokan Anandarajan, and Cadambi A. Srinivasan, eds. Business intelligence techniques: a perspective from accounting and finance. Springer Science & Business Media, 2012.
2. Michalewicz, Zbigniew, et al. Adaptive business intelligence. Springer Berlin Heidelberg,2006.

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Session Plan:(*Each session consists of 60 minutes*)

Session No.	Topics to be covered
1-6	**Introduction to Analytics and Business Analytics** Introduction to Analytics and Business Analytics Foundations of Analytics, Historical evolution of analytics, Contemporary significance of analytics, Future trends in analytics, Business Analytics Fundamentals, Definition and importance of business analytics, Types of analytics and their applications, Role of predictive analytics in decision-making, Analytics Trends and Predictive Enterprises, Analysis of past, present, and future analytics trends, Transition towards predictive enterprise models, Case studies showcasing successful analytics implementations
7-14	**Business Intelligence and Analytics** Explaining what is analytics, defining various types of analytics, Demonstrating how to apply analytics, Describing business intelligence, Demonstrating how to apply business intelligence, Learning how to access content, use reports, and create dashboards, Learning how personalize the IBM Cognos Analytics portal, Group, format, and sort list reports, Describing the various options for aggregating data ,Creating a multi-fact query ,Creating a report with repeated data, Creating filters to narrow the focus of reports ,Examining detail filters and summary filters ,Determining when to apply filters on aggregate data, Formatting and sort crosstab reports , Creating complex crosstab reports using drag and drop functionality, creating crosstab reports using unrelated data items, creating charts containing peer and nested columns, presenting data using different chart type options, adding context to charts, Creating and reuse custom chart palettes
15-21	**IBM Cognos Analytics** Author Reports Advanced & Active Reports: create reports based on query relationships, create advanced dynamic reports, Design effective prompts, create additional advanced reports, Examine the report specification, distribute reports through bursting, Enhance user interaction with HTML.
22-26	**Advanced Query Modeling and Report Layout Design with SQL Customization** Building query models and connect them to the report layout Editing an SQL statement to author custom queries, Adding filters and prompts to a report using the query model, Creating reports by merging query results, Creating reports by joining queries, Combining data containers based on relationships from different queries, Filtering reports on session parameter values, Navigating a briefing book using a table of contents, Creating dynamic headers and titles that reflect report data, Navigating to specific locations in reports, Creating a customer invoice report, Controlling report displays using prompts, Specifying conditional formatting values using prompts, Specifying conditional rendering of objects based on prompt selection, Creating sorted and filtered reports based on prompt selection, Creating a report that displays summarized data before detailed data, Highlighting alternate rows in a list report ,Creating a report using an external data file
27-30	**IBM Cognos Active Reports** Introduction to IBM Cognos Active Reports, Use Active Report connections, Active Report charts, visualizations and decks. Project: Design and Develop an Interactive Active Report Dashboard for Sales Analysis, Implementing Dynamic Data Visualization in Active Reports for Financial Performance Analysis

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Gautam Buddha University
School of Management
Greater Noida
MBA (Business Analytics and Data Science) Programme

Course: Data Analytics using R (Practical)

Code:	MBA-452	**Year/Sem:**	**2024**-2025/II
Department:	School of Management	**Credits :**	02
Sessions:	30	**Each Session:**	**1 2 0** Minutes

Course Objectives

1. Understanding Data Analysis
2. To learn R as a Programming Language.
3. Able to do visualization with R
4. Implementing R for statistical analysis
5. Implementing R for prescriptive analysis

Course Outcomes

At the end of the course the students should be able to:

1. How to analyze the data
2. Able to use R for Decision making
3. Able ro do any visualization with R
4. Ability to apply statistical techniques using R
5. Act like a data analyst

Pedagogy:

This pedagogy will make use of analytical and computer programs using R and SPSS Modeler having applications mostly in business areas. The sessions will be conducted in Computer lab where students are expected to implement the programs in order to develop programming and problem-solving skills.

Evaluation Scheme: (Practical Paper)

Assignments / Class participation, Machine Test and Presentations	: 50%
End-Sem Exam (Lab Exam)	: 50%

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Session Plan:

Unit	Session (hrs)	Topics to for Covered
Unit-I Introduction to data analysis:	1-5	Overview of Data Analytics, Need of Data Analytics, Nature of Data, Classification of Data, Structured, Semi- Structured, unstructured, Characteristics of Data, Application of Data Analytics.
Unit-II R Programming Basics:	6-11	Overview of R programming, Environment setup with R Studio, R Commands, Variables and Data Types, Control Structure, Array, Matrix, Vectors, Factors, Functions, R packages.
Unit-III Data Visualization Using R	12-18	Reading and getting data into R (External Data) using CSV files, Web Data, JSON files, Databases, Excel files. Working with R Charts and Graphs: Histograms, Box plots, Bar Charts, Line Graphs, Scatter plots, Pie Charts,
Unit-IV Statistics with R	19-24	Random forest, Decision Tree, Normal and Binomial distribution, Time Series Analysis Linear and Multiple Regression, Logistic Regression, Survival Analysis.
Unit-V Prescriptive Analytics	25-30	Creating Data for Analytics through designed experiment, Creating Data for Analytics through active learning, Creating Data for Analytics through reinforcement learning.

Text Books:

An Introduction to R,Notes on R, A Programming Environment for data analysis and Graphics, W.N. Venables, D.M.Smith and the R Development core Team Version 3.0.1 (2013-05-16) URL:https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf

Reference Books:

1. Jared P Lander, R for everyone: advanced analytics and graphics, Pearson Education.
2. Dunlop, Dorothy D,and Aajit C.Tamhane. Statistics and data analysis: form elementary to intermediate, Prentice Hall.
3. G Casella and R.L Berger, Statistics Inference, Thomson Learning 2002
4. P Dalgaard introductory Statics with R, Springer.
5. Michael Berthold David j. Hand Intelligent Data analysis, Springer.
6. Montgomery, Douglas C. and George C. Runger, Applied statistics and probability for engineers.John Wiley & Sons.
7. Joeph F hair, William C Black etal, ''Multivariate Data Analysis'' Pearson Education.
8. Mark Gardener, ''Beginning R-the statistical programming Language'' John Wiley & Sons.
9. W.N.Venables, D.M Smith and the R Core Team, "An Introduction to R".

Note: Latest edition of the books to be followed.

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Gautam Buddha University
School of Management
Greater Noida

MBA Programme

Course: Business Communication Lab **MB- 470**

Faculty:	**Dr. Manjri Suman**	**Year/Sem:**	2024-25/II
Department:	Management	**Credit:**	02
Sessions:	30	**Each Session:**	120 Minutes

Introduction to the Course:

According to the Webster's New World College definition, "A language laboratory is a classroom in which students learning a foreign language can practice sound and word patterns individually or under supervision with the aid of audio equipment, etc."

Communication skills have a huge role in the portrayal of an individual's personality. Thus, having a decent language proficiency is imperative if one wants to have a successful career. A language lab is a place which improves the communication skills of students in a second or foreign language. Since the language of business and professional communication is English in India, this paper exclusively focuses on good communication skills with a special emphasis on English Proficiency.

Course Objectives:

Research carried out by the Carnegie Institute of Technology shows that 85 percent of your financial success is due to skills in "human engineering," your personality and ability to communicate, negotiate, and lead. Shockingly, only 15 percent is due to technical knowledge.

Keld Jensen - Forbes Magazine.

Since English is the language of professional communication in India, the lab will increase the English language proficiency of the students. Many students have a sound technical knowledge of their respective area but unfortunately, they lack in communication skills. The language lab is designed for MBA students as it will focus on the following aspects:

i. Hone their communication skills (both verbal and non-verbal).
ii. Make them fluent in speaking English.
iii. Enhance their listening power as well.
iv. Prepare them to face interviews, confidently.
v. Improve their business writing skills i.e. writing skills in the professional domain.

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Course Outcomes:

This course covers the practical application of language. It covers all the four crucial aspects of language learning i.e. Listening, Speaking, Reading and Writing.

i. Students will be engaged in practicing all these aspects of language through active participation in class.
ii. They will get a good practice of speaking skills which has comparatively narrow scope in other courses.
iii. They will improve their professional writing skills, for example letters and emails etc.
iv. They will get comfortable with everyday business communications in English.
v. At the end of course the students turn out to be confident in speaking at public platforms or in formal situations.

Pedagogy:

This course covers the practical application of language for effective communication. It covers all the four crucial aspects of language: Listening, Speaking, Reading and Writing. Students will be engaged in practicing all these aspects of language through active participation in class. Constant feedback in the form of suggestions will help them incorporate changes in their speech, writing, and professional language as a whole.

Session Plan:

Session 1: Basic Introduction Exercise. Basic components of self-introduction will be taught that students can incorporate while introducing themselves. Suggestions will be provided including grammatical corrections and better sentence structures.

Session 2: Self Introduction. After incorporating the suggestions.

Session 3-4: Group Discussion. Do's and Don'ts of GD. How to be emphatic contributions. This session will focus on building confidence and enhancing public speaking skills.

Session 5: Writing Practice. Components and Format of Resumé. Teaching them how to prepare an impressive resumé, business reports and business proposal. At the end of the session, students will be given an assignment to prepare a formal resumé.

Session 6: Writing Practice. Discuss upon the resumé prepared by the students.

Session 7-8: Understanding English pronunciation. Articulatory Phonetics.

Session 9: Reading session focused on various case studies. Their reading comprehension skill will be the agenda of the session.

Session 10: Group Discussion. Working on fluency and public speaking. The idea is to develop confidence through bilingual interactions first. The students will gradually move towards using solely English sentences.

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Session 11: Pronunciation practice. Phonics, Voice modulation, Accent and Articulation.

Session 12-13: Writing session. How to write a job application and a cover letter to go with the job application.

Session 14-15: Listening exercise. Audio links will be provided to students to enhance their listening practice.

Session 16-17: Oral Presentations skills. Students will be asked to prepare a PowerPoint on a related topic and give a presentation on it. This will further enhance their public speaking skill and help in removing their stage fear.

Session 18: Writing session. Practice of email writing. Email etiquettes will also be taught.

Session 19-20: Listening Exercise. Students would hear model audio to acquaint them with listening English sentences and everyday conversations in English. They will also be shown videos of mock interview for listening and speaking practice.

Session 21: Reading session. Reading comprehension and forming answers.

Session 22: Letter writing practice. Part 2

Session 23: Pronunciation. Word stress and syllable structure. Understanding English pronunciation. A focused session on these aspects will improvise their pronunciation skills.

Session 24: Interview Practice with a special focus on Body language.

Session 25-26: Mock interview. Students will be asked to give a proper introduction of themselves again after incorporating the suggestion given in the 1st session. Dressing sense, language, politeness, appropriate body language will all be a part of assessment and training.

Session 27-28: Self-evaluation for students. Students record their own voice and play back the recordings, interact with each other and the teacher to understand their drawbacks and areas of improvisation. This collaborative nature usually increases students' motivation and their learning experience

Session 29-30: Group Discussion. By now students will have good understanding of proper body language and how to go about a GD. Making small and content rich contributions are crucial to make an impact on the examiner/interviewer. This session will be focused on honing such skills.

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Books/ Web References/ Software Tools:

Relevant material will be provided in class. Providing the material and links beforehand will discourage students from actually coming to the class and they would engage in passive learning. Since this is a lab class, active learning is crucial for their skills and personality development. However, few books and web links are mentioned here for reference.

Suggested Readings and Links:

i. Business Correspondence and Report Writing, R.C. Sharma & Krishna Mohan (Latest Edition).
ii. Basic Business Communication Skills for Empowering the Internet Generation, Laskar & Flatley, Latest Edition).
iii. Negotiations, Michael L. Spangle & Myra Warren Isenhart, Sage Publications, South Asia Edition.
iv. Business Communication, Meenakshi Raman & Prakash Singh, Oxford University Press
v. Web links:
https://ielts-up.com/listening/ielts-listening-practice.html
https://takeielts.britishcouncil.org/take-ielts/prepare/free-ielts-english-practice-tests/listening

Note: Latest edition of all the books to be followed.

Evaluation Scheme:

The students will be evaluated throughout the course for the class performance and engagement, presence of mind, curiosity to learn, and application of the teachings in their personality development. Just showing up on the final assessment day will not suffice for their respective grades.

They will also be required to maintain a lab record in the form of note book which will track the learnings of all the sessions. Written assignments would be maintained there. Even oral sessions would be recorded by noting down the key learnings of the class.

The overall assessment would be divided into the following components:

i. End semester exam (written and oral) – 50%
ii. Class participation, interaction and involvement – 20%
iii. Lab Record – 20%
iv. Improvement over time in their speaking and writing skill. Including the desire to learn and grow. 10%

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Gautam Buddha University
School of Management
Greater Noida

TEACHING-LEARNING PLAN
MBA Programme

Course: Business Communication **MB- 468**

Faculty:	**Dr. Manjri Suman**	**Year/Sem:**	2024-25/II
Department:	Management	**Credit:**	02
Sessions:	30	**Each Session:**	60 Minutes

Introduction to the Course:

Communication for Managers is intended to help the learners think strategically about communication and to aid in improving writing, presentation, and interpersonal communication skills within a managerial setting. The value of communication, particularly in business situations is indispensable. In this highly competitive business environment, effective skills of communication are required to put one's ideas across efficiently. The course intends to enhance quality of interaction at personal, interpersonal and team levels for the fulfillment of organizational goals and objectives.

Course Objectives:

i. To provide the learners with the basic concepts of Communication.
ii. To acquaint them with the nuances of Communication in the business organizations.
iii. To equip them with desired skills of communication to handle multi-tasked contexts.

Course Outcomes:

After successful completion of the course the learners will be able to:

i. Relate to and handle nuances of communication effectively in the contemporary context.
ii. Connect to and communicate in business environments as professionals.
iii. Understand the role and relevance of communication, with special reference to the domains they will be working in.

Pedagogy:

Writing and speaking skills necessary for a career in management. Students polish communication strategies and methods through discussion, example-based learning, and practice. Several written and oral assignments, mostly based on material from other subjects and from career development activities will be discussed. The teaching

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methodology will be a judicious blend of lectures, demonstrations, technical presentations, class works, real life situation analysis, case study, presentation by students of their project work, and exercises. The content delivery shall be in a highly interactive mode, expected to be participated by all, and therefore students are advised to come equipped with sufficient readings on the respective topics. Cases will be intimated later. Exercises and materials/handouts will be provided at the lecture sessions.

Session Plan: Each session of 60 minutes.

Session 1. Communication: Foundation and Analysis

Session 2-3. Verbal and Non-Verbal Communication

Session 4. Communicating within Organization

Session 5-6. Barriers to Communication

Session 7. Gateways to Effective Communication

Session 8. Group Communication: Characteristics of Effective Groups, From Groups to Teams, Meeting Management

Session 9. Case Presentations

Session 10. Attributes to Effective Communication

Session 11. Language Skills: LSRW

Session 12. Strategies of Effective Listening

Session 13. Strategies for Effective Reading

Session 14. Strategies for Effective Writing

Session 15-16. Strategies for Effective Speaking: Designing and Delivering Oral Presentation

Session 17. Case Presentations

Session 18. Communicating Electronically

Session 19-20. Report Writing

Session 21-22. Business Proposals

Session 23. Business Letters

Session 24. Communication for Employment: Preparing Résumés and Application Messages

Session 25-26. Notices, Office Memos, Minutes, Tender Notices and other Correspondence

Session 27. Communication and Negotiations, Nature of Negotiations, Process of Negotiations

Session 28. Persuasive Communication

Session 29-30. Oral Presentations and Project submission.

Text Book:

i. Business Communication. Meenakshi Raman & Prakash Singh, Oxford University Press
ii. Business Communication Today. Courtland L. Bovee, John V. Thill, Roshan Lal Raina. India: Pearson India.

Reference Books:

i. BCOM, Lehman Carol M., Dufrene Debbie D., Sinha Mala, Cengage Learning Publication.
ii. Business Correspondence and Report Writing, R.C. Sharma & Krishna Mohan, TMH.
iii. Basic Business Communication Skills for Empowering the Internet Generation, Lasikar & Flatley, TMH.
iv. Contemporary Business Communication, Scot Ober, Biztantra, (Indian Adaptation).
v. Negotiations, Michael L. Spangle & Myra Warren Isenhart, Sage Publications.

Note: Latest edition of all the books to be followed.

Evaluation Scheme: Evaluation Schemes include submission and presentation of Class Participation, Projects and Assignments, Case-Studies and Presentations, and Mid-Semester & End-Semester Tests. It is essential to follow the deadlines for preparation and submission of assignments, projects, and presentations as indicated by the course coordinator.

i. End semester exam (written) – 60%
ii. Mid semester – 25%
iii. Internal Assessment 15%

    a. Class participation and Presentation – 07%
    b. Projects & Assignments- 08%

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Class participation: The attendance requirement for successful completion of the course is a minimum of 75%. Students are encouraged to participate actively and contribute to the quality of teaching-learning process.

Projects & Assignments: Projects and Assignment shall be comprehensive in nature. It is mandatory that the participants use Power Point Slides in their class presentations. Credit shall be given for structured presentation, analytical content and ability to respond to queries.

Case studies & presentations: Cases shall engage students in exploring current communication challenges in the business environment. Students are expected to do little research before presenting final outcomes of the case in oral and written formats. Mid-Sem Exam: Mid-Sem test shall comprise of medium-size answer questions. The test shall be based broadly on the syllabus covered till mid-term teaching.

End-Sem Exam: End-Sem test shall comprise of fundamental, analytical and comprehensive questions. The test shall be based on the entire course coverage (more weightage would be given to post mid-term syllabus) and designed to test the conceptual clarity of the subject and their applications.

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Gautam Buddha University
School of Management
Greater Noida
Master in Business Administration
(Business Analytics and Data Science)

Course: Data Base Management System Lab (Practical)

Code:	MBA-474	**Year/Sem:**	2024-2025/II
Department:	School of Management	**Credit:**	02
Sessions:	30	**Each Session:**	120 Minutes

Laboratory Overview

This course introduces the core principles sand techniques required in the design and implementation of database systems.

This introductory application-oriented course covers the relational database systems RDBMS -the predominant system for business scientific and engineering applications at present. It includes Entity-Relational model, Normalization, Relational model, Relational algebra, and data access queries as well as an introduction to SQL. It also covers essential DBMS concepts such as: Transaction Processing, Concurrency Control and Recovery. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in information technology applications.

Course Objectives:

1. To explain basic data base concepts, applications, data models, schema sand instances.
2. To demonstrate the use of constraints and relation a algebra operations.
3. **Describe** the basics of SQL and construct queries using SQL.
4. To emphases size the importance of normalization in data bases.
5. To facilitate students in Data base design
6. To familiarize issues of on currency control and transaction management.

Course Outcome:

At the end of the course the student shareable to:

1. Apply the basic concepts of Data base Systems and Applications.
2. UsethebasicsofSQLandconstructqueriesusingSQLindatabasecreationandinteraction.
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4. Analyze and Select storage and recovery techniques of data base system.

Pedagogy:

This pedagogy will make use of analytical and computer programs using Python and SPSS Modeler having applications mostly in business areas. The sessions will be conducted in Computer lab where students are expected to implement the programs in order to develop programming and problem-solving skills.

Evaluation Scheme: (Practical Paper)

Assignments / Class participation, Machine Test and Presentations	: 50%
End-Sem Exam (Lab Exam)	: 50%

List of Experiments:

Experiment-1
Student should decide encase study and for mutate the problem statement.

Experiment-2
Conceptual Designing using ER Diagrams (identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.)
Note: A Student is required to submit a document by drawing ER Diagram to the Lab teacher.

Experiment-3
Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, represent attributes as columns, identifying keys)
Note: Students are required to submit a document showing the database tables created from ER Model.

Experiment-4
Normalization-
To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form

Experiment-5
Creation of Tables using SQL- Overview of using SQL tool, Data types in SQL, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables

Experiment-6
Practicing DML commands-Insert, Select, Update, Delete

Experiment-7
Practicing Queries using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSECT, CONSTRAINTS, etc.

Experiment-8
Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equip).

Experiment-9

Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping.

Experiment-10

Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger

Experiment-11

Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure.

Experiment-12

Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

Experiment-13

Design and implement a database schema for company data base, banking data base, library information system, payroll processing system, student information system.

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Gautam Buddha University
School of Management
Greater Noida

MBA (Business Analytics and Data Science) Programme

Course: Database Management System	**Code: MBA-470**
Session/Sem: 2024-25/II	Credits: 02
Department: Business Management	**Sessions:** 30
Batch: 2024-25	**Each Session:**60 Minutes

COURSE OVERVIEW:

This course introduces the core principles and techniques required in the design and implementation of database systems. This introductory application-oriented course covers the relational database systems RDBMS - the predominant system for business, scientific and engineering applications at present. It includes Entity-Relational model, Normalization, Relational model, Relational algebra, and data access queries as well as an introduction to SQL. It also covers essential DBMS concepts such as: Transaction Processing, Concurrency Control and Recovery. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in information technology applications.

COURSE OBJECTIVES:

1. To Teach the basic database concepts, applications, data models, schemas and instances.
2. To familiarize Entity Relationship model for a database.
3. To Demonstrate the use of constraints and relational algebra operations.
4. To Describe the basics of SQL and construct queries using SQL.
5. To Emphasize the importance of normalization in databases.
6. To Demonstrate the basic concepts of transaction processing and concurrency control.
7. To familiarize the concepts of database storage structures and identify the access techniques.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Use the basic concepts of Database Systems in Database design
2. Apply SQL queries to interact with Database
3. Design a Database using ER Modelling
4. Apply normalization on database design to eliminate anomalies
5. Analyze database transactions and can control them by applying ACID properties.

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UNIT – V (Sessions: 23-30)

TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability.

CONCURRENCY CONTROL AND RECOVERY SYSTEM: Concurrency control, lock based protocols, time-stamp based protocols, validation based protocols, multiple granularity. Recovery system - failure classification, storage structure, recovery and atomicity, log- based recovery, shadow paging, buffer management, failure with loss of non-volatile storage, advanced recovery techniques, remote backup systems.

OVERVIEW OF STORAGE AND INDEXING: Tree structured indexing - intuition for tree indexes, indexed sequential access method (ISAM), B+ Trees - a dynamic tree structure.

TEXT BOOKS:

1. Raghurama Krishnan, Johannes Gehrke , Database Management Systems, Tata McGraw Hill, New Delhi, India.
2. Elmasri Navate, Fundamentals of Database Systems, Pearson Education, India.

REFERENCE BOOKS:

1. Abraham Silberschatz, Henry F. Korth, S. Sudarshan (2005), Database System Concepts, McGraw-Hill, New Delhi, India.
2. Peter Rob, Carlos Coronel, Database Systems Design, Implementation and Management.

Note: Latest edition of books to be followed.

Sanjay

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